

21/7/25

Study of Different Types of Network Cables.

Ex: 2

Aim:

Study of different types of Network cables.

a) understand different types of network cable

Different type of cable used in network are:

1. Unshielded Twisted Pair (UTP) cable.

2. Shielded Twisted Pair (STP) cable.

3. Coaxial cable.

4. Fibre optic cable.

Cable type	Category	Maximum Data transmission	Advantages/Disadvantages	Application/Use
UTP	category 3	10 bp	Advantages + cheaper in cost * Easy to install as they have a smaller overall diameter	10Base-T Ethernet Fast Ethernet Gigabit Ethernet Fast Ethernet Gigabit Ethernet
	category 5	up to 100 mbps.		
	category 5e	10 Gbps	Disadvantage + More prone to (EMI) Electron interference and noise	
STP	category 6, 6a	10 Gbps.	Advantages * Shielded * Easier than UTP * Less Susceptible to noise and Interference.	Gigabit Ethernet 10G Ethernet (SSM) widely used in data centres
SSTP	Category 7	10 Gbps	Disadvantage * Expensive * Greater	Gigabit

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	* Immune to interference * Low loss bandwidth * Versatile	Speed of Signal is 500m, Tolerant network Speed, and connection
			Disadvantages * Limited distance * Cost * Size is bulky	
Fibre optic cable	Single mode Multimode	100 Gbps	Advantages * High Speed * High bandwidth * High Security * Long distance	* Maximum distance of fibre optic cable is 100 meters
			Disadvantages * Expensive * Requires skilled installers	

D Make your own Ethernet cross-over cable / Straight cable.

Tools and Parts needed:

* Ethernet Cabling, CAT 5e is certified for gigabit support, but CAT 5 cabling works as well just over shorter distances.

* A crimping tool. This is an all in one network tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.

* Two RJ45 Plugs.

* Optional two Plug Shields.

Step 1: To start construction of the device begin by threading shield into the cables.

Step 2: Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3: After, you will need to untangle the wires. There should be four "twisted pairs" referring back to the select, arranged them from top to bottom. one end should be in arrangement A and the other in B.

Step 4: Once the order is correct, bunch them together in a line, and if there are many that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 Plug without messing up the order. To do so, hold the Plug with the clip side facing away from you and have the gold pins facing toward you as shown.

Step 5: Next, Push the cable right in. The notch at the end of the Plug needs to be just over the cable shielding, and if it isn't that means that you stripped off too much shielding. Simply snip the cable back a little more.

Step 6: After the wires are securely sitting inside the Plug, insert it into the crimping tool and Push down.
It should be shaped correctly, but Pushing too hard can crack the fragile Plastic Plug.

Step 7: Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cables).

Student Observation

1) Which command is used to find the reachability of a host machine from your devices?

Ans: `Ping <hostname or IP>`
`Ping Google.Com`

2) Which command will give the details of hops taken by Packet to reach its destination?

`tracert <hostname>`
`tracert google.com`

3) Which command displays the IP configuration of your machine.

Ans: `ipconfig` shows all the configuration details of your machine.

4) Which command displays the TCP Port status?

`netstat`
`netstat -s`

5) Write the command to modify the IP configuration in a Linux machine?

Ans: to add IP address `ifconfig eth0 192.168.1.254/24 dev eth0 netmask 255.255.255.0`
to add IP address `ifconfig eth0 192.168.1.254/24 dev eth0 netmask 255.255.255.0`

Result:

Thus the behaviour of network devices has been successfully analysed using CISCO Packet Tracer Simulator.